

Data Quality Report — Collections Dataset

Every issue below was **detected**, **quantified**, and given a **treatment rule** that is implemented in `golden/build_golden.py`. Issues are ordered by business impact.

Summary: Raw → Rejected/Corrected → Golden

Table	Raw rows	Action	Golden rows
payments	25,500	drop 500 exact dup rows; keep in-window	25,000 events / 17,534 recovery (SUCCESS)
agents	30,000	resolve to <code>agent_id</code> (latest <code>updated_at</code>)	1,000 agents
accounts	30,000	de-dup on <code>account_id</code>	30,000
call_dispositions	35,000	standardise PTP → <code>PROMISE_TO_PAY</code>	35,000
Recovery total	₹1.341 Cr gross	remove dup inflation	₹131.6 Cr clean

A. Duplicate payments — CONFIRMED (two different problems, opposite treatments)

Detection. `payment_id` had 25,500 rows but 25,000 unique IDs → 500 IDs appear twice; 97% of those pairs are byte-identical across all columns → classic ingestion retries. **Treatment.** De-duplicate on `payment_id` (keep first). Removes 500 rows / **₹2.5 Cr of phantom recovery (~2%)**. **Trap avoided.** `payment_reference` looks like a natural key but is **reused across 3,407 references spanning multiple accounts, amounts and dates**. De-duplicating on it would have destroyed **~17% of legitimate recovery**. Rule: *never* de-dup payments on `payment_reference`.

B. Attribution error — CONFIRMED (structural)

Detection. Payments carry no `campaign_id` or `call_id`; any channel/campaign attribution must be inferred (last-touch). We tested whether contact even predicts payment and found it does not (see §Statistical). Therefore last-touch attribution assigns credit to whichever interaction happened to be most recent, not to a cause. **Treatment.** Do not attribute recovery to campaign/channel/agent in the golden layer. Report recovery at account and portfolio level; treat any channel-level “conversion” as correlational only.

C. Timezone problems — CONFIRMED

Detection. `accounts`, `calls`, `agent_sessions`, `vendor_telephony` declare three zones (UTC, Asia/Kolkata +5:30, Asia/Dubai +4), but `event_at` is a **naive timestamp**: mean call hour is 11.5 for *all three* zones, ~25% of calls fall in “00:00–05:59”, and 30,191 `account_status_history` rows are recorded *at before* `event_at` by ≤ 1 day (an offset artifact). **Treatment.** Keep `event_at` for daily/monthly volume (robust to ≤ 5.5 h shifts). Expose a best-effort UTC column, and **flag intraday (calling-hour) analysis as unreliable** — proper local-time reconstruction would move ~half of events across hour boundaries.

D. Vendor / disposition code changes — CONFIRMED (synonyms, not a clean cutover)

Detection. `call_dispositions.disposition_version` $\in \{\text{legacy}, \text{v1}, \text{v2}\}$. Codes PTP and `PROMISE_TO_PAY` both appear in **all three** versions at similar volume → they are synonyms coexisting, not a versioned rename. **Treatment.** Standardise PTP → `PROMISE_TO_PAY` before computing PTP/contact metrics. `vendor_telephony` mappings were checked and are stable across the window.

E. Agent identity — CONFIRMED (severe)

Detection. `agents` has 30,000 rows but only **1,000 distinct** `agent_id`. `employee_code` is not an identity: `EMP000001` maps to 30 different `agent_id`s with different names, vendors and teams. Within a single `agent_id`, `name/vendor/team/joined_at` all conflict. **Treatment.** `agent_id` is the only trustworthy key; resolve the dimension to the latest `updated_at` per `agent_id`. **Consequence:** “agent tenure” cannot be computed reliably (`joined_at` is corrupted), so agent-tenure conclusions are withheld.

F. Portfolio mix change — CHECKED, NOT PRESENT

Risk segment, loan type and DPD are near-uniform and **stable across all months**; monthly targeted volume is flat (~5,600/mo). No evidence the business acquired a fundamentally different portfolio. So mix is *not* an explanation for any apparent change.

G. Denominator manipulation — CHECKED, NOT PRESENT

Targeted-account counts and `account_status_history` events are stable month to month; no cohort of unsuccessful accounts silently drops out of the population. Conversion denominators are intact.

Other issues

- **Out-of-window leakage:** a stray `2025-12` call record; excluded by the window filter.
- **Payment status mix:** 70% SUCCESS, 15% FAILED, 10% PENDING, 5% REVERSED. Only SUCCESS is recovery.
- `outstanding > principal` on 43% of accounts — plausible (accrued interest/fees), retained.

Statistical robustness check (why the “cleaning” conclusions hold)

- RPC vs no-RPC payment rate: 44.7% vs 43.9%, χ^2 $p = 0.19$ (not significant).
- Attempts vs payment: point-biserial $r = 0.006$, $p = 0.30$ (no dose-response).
- Recovery Jan vs Jul: +0.01%. Full-month recovery CV: 4.1%.

These are transparent, reproducible tests (no black-box models), consistent with the assignment’s preference for explainable methods.